# SEPARATION AXIOMS ARE STABLE UNDER COMPOSITION, BASE CHANGE AND PRODUCTS

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 4518-4551, following the statement at 4505-4516.

*Editorial note.* The phrase 'which is easily seen to be the base change of Delta_f' (~4539) is a diagram-anchored identification reading; the displayed formula fixes the comparison; same precedent as cards L26341/L26383.

*Allowed standard prerequisites (editorial).*

- Fibre products after a morphism stay closed/surjective. lemma-fibre-product-after-map (same chapter, cited at ~4525). The second arrow X x_Y X -> X x_Z X is closed.
- Base change preserves immersions. lemma-fibre-product-immersion (same chapter, cited at ~4543). Closed immersions stable under base change.
- Base change preserves quasi-compactness. lemma-quasi-compact-preserved-base-change (same chapter, cited at ~4544). Quasi-compact morphisms stable under base change.

*Source reference.* Original lemma 01KR: sources/stacks-schemes/pinned-original-source.tex, source lines 4455–4455. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 01JU: sources/stacks-schemes/pinned-original-source.tex, source lines 3547–3547. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 01K5: sources/stacks-schemes/pinned-original-source.tex, source lines 3843–3843. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Permanence properties.*

(1) *A composition of separated morphisms is separated.*
(2) *A composition of quasi-separated morphisms is quasi-separated.*
(3) *The base change of a separated morphism is separated.*
(4) *The base change of a quasi-separated morphism is quasi-separated.*

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*Date*: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(5) *A (fibre) product of separated morphisms is separated.*
(6) *A (fibre) product of quasi-separated morphisms is quasi-separated.*

*Proof.* Let $X \to Y \to Z$ be morphisms. Assume that $X \to Y$ and $Y \to Z$ are separated. The composition

$$X \to X \times_Y X \to X \times_Z X$$

is closed because the first one is by assumption and the second one by Lemma 01KR. The same argument works for "quasi-separated" (with the same references).

Let $f : X \to Y$ be a morphism of schemes over a base $S$. Let $S' \to S$ be a morphism of schemes. Let $f' : X_{S'} \to Y_{S'}$ be the base change of $f$. Then the diagonal morphism of $f'$ is a morphism

$$\Delta_{f'} : X_{S'} = S' \times_S X \longrightarrow X_{S'} \times_{Y_{S'}} X_{S'} = S' \times_S (X \times_Y X)$$

which is easily seen to be the base change of $\Delta_f$. Thus (3) and (4) follow from the fact that closed immersions and quasi-compact morphisms are preserved under arbitrary base change (Lemmas 01JU and 01K5).

If $f : X \to Y$ and $g : U \to V$ are morphisms of schemes over a base $S$, then $f \times g$ is the composition of $X \times_S U \to X \times_S V$ (a base change of $g$) and $X \times_S V \to Y \times_S V$ (a base change of $f$). Hence (5) and (6) follow from (1) – (4). □